

# Lasso Regression Summary: 2020 Q1

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED] | 0.1958      | [0.0000, 0.6209]  | 0.7820  | 0.1815    |
| Mobile Weight (grams)             | 0.0129      | [-0.0839, 0.1265] | 0.2520  | 0.0931    |
| RAM Memory (GB) [SELECTED]        | 0.2977      | [0.0000, 0.6739]  | 0.9280  | 0.1728    |
| Front Camera (MP) [SELECTED]      | 0.1738      | [0.0000, 0.3031]  | 0.9640  | 0.0708    |
| Max Back Camera MP [SELECTED]     | -0.1248     | [-0.4091, 0.0000] | 0.6280  | 0.1254    |
| Number of Cameras [SELECTED]      | 0.2335***   | [0.0874, 0.4013]  | 0.9990  | 0.0809    |
| Processor Level (encoded)         | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Battery Capacity (mAh)            | -0.0689     | [-0.4424, 0.2424] | 0.5920  | 0.1867    |
| Screen Size (inches) [SELECTED]   | 0.0982      | [-0.0942, 0.3064] | 0.7700  | 0.1171    |

Model Information:

- Sample Size: 21
- R<sup>2</sup> Score: 0.9175
- Optimal Alpha: 0.001397
- Features Selected: 6/9

Lasso Regression Summary: 2020 Q2

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0)            | 0.0010      | [0.0000, 0.0046]  | 0.0270  | 0.0084    |
| Mobile Weight (grams) [SELECTED]  | 0.0370      | [-0.0043, 0.1392] | 0.7890  | 0.0364    |
| RAM Memory (GB)                   | 0.0134      | [0.0000, 0.1745]  | 0.1330  | 0.0443    |
| Front Camera (MP) [SELECTED]      | 0.1112      | [0.0000, 0.1757]  | 0.9400  | 0.0403    |
| Max Back Camera MP [SELECTED]     | -0.0842     | [-0.1658, 0.0000] | 0.6580  | 0.0648    |
| Number of Cameras [SELECTED]      | 0.1402***   | [0.0768, 0.2249]  | 0.9980  | 0.0371    |
| Processor Level (encoded)         | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Battery Capacity (mAh) [SELECTED] | 0.0291      | [-0.1609, 0.0953] | 0.7940  | 0.0588    |
| Screen Size (inches)              | 0.0096      | [0.0000, 0.1352]  | 0.0910  | 0.0359    |

Model Information:

- Sample Size: 22
- R² Score: 0.8957
- Optimal Alpha: 0.011650
- Features Selected: 5/9

Lasso Regression Summary: 2020 Q3

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0)            | 0.0129      | [0.0000, 0.2051]  | 0.0710  | 0.0586    |
| Mobile Weight (grams)             | 0.0118      | [-0.0022, 0.0722] | 0.3970  | 0.0286    |
| RAM Memory (GB)                   | 0.0006      | [0.0000, 0.0000]  | 0.0280  | 0.0277    |
| Front Camera (MP)                 | 0.0211      | [-0.1303, 0.1599] | 0.5240  | 0.0678    |
| Max Back Camera MP                | -0.0044     | [-0.3026, 0.0785] | 0.5330  | 0.0951    |
| Number of Cameras [SELECTED]      | 0.1358***   | [0.0461, 0.2351]  | 0.9980  | 0.0492    |
| Processor Level (encoded)         | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Battery Capacity (mAh) [SELECTED] | 0.0896      | [0.0000, 0.2518]  | 0.9140  | 0.0760    |
| Screen Size (inches)              | 0.0014      | [0.0000, 0.0000]  | 0.0080  | 0.0210    |

Model Information:

- Sample Size: 24
- R² Score: 0.5908
- Optimal Alpha: 0.025959
- Features Selected: 2/9

Lasso Regression Summary: 2020 Q4

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0)            | 0.0317      | [0.0000, 0.3248]  | 0.1560  | 0.0915    |
| Mobile Weight (grams)             | 0.0187      | [-0.0032, 0.0891] | 0.5040  | 0.0418    |
| RAM Memory (GB)                   | -0.0011     | [0.0000, 0.0000]  | 0.0140  | 0.0364    |
| Front Camera (MP)                 | 0.0282      | [-0.1616, 0.1810] | 0.5730  | 0.0823    |
| Max Back Camera MP [SELECTED]     | 0.0268      | [-0.3392, 0.1266] | 0.7240  | 0.1215    |
| Number of Cameras [SELECTED]      | 0.1625***   | [0.0616, 0.2968]  | 0.9990  | 0.0608    |
| Processor Level (encoded)         | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Battery Capacity (mAh) [SELECTED] | 0.0762      | [-0.0051, 0.3095] | 0.8150  | 0.0967    |
| Screen Size (inches)              | 0.0036      | [0.0000, 0.0000]  | 0.0220  | 0.0286    |

Model Information:

- Sample Size: 24
- R² Score: 0.5934
- Optimal Alpha: 0.026642
- Features Selected: 3/9

| Feature                   | Coefficient | 95% CI            | P-Value | Std Error |
|---------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0)    | 0.0049      | [0.0000, 0.1183]  | 0.0310  | 0.0284    |
| Mobile Weight (grams)     | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| RAM Memory (GB)           | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Front Camera (MP)         | 0.0124      | [0.0000, 0.2569]  | 0.0460  | 0.0580    |
| Max Back Camera MP        | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Number of Cameras         | 0.1314      | [0.0000, 0.4120]  | 0.4750  | 0.1461    |
| Processor Level (encoded) | -0.0352     | [-0.1568, 0.0000] | 0.2610  | 0.0599    |
| Battery Capacity (mAh)    | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |
| Screen Size (inches)      | 0.0000      | [0.0000, 0.0000]  | 0.0000  | 0.0000    |

Model Information:

- Sample Size: 37
- R² Score: 0.0000
- Optimal Alpha: 0.230929
- Features Selected: 0/9

# Lasso Regression Summary: 2021 Q2

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.9720***   | [0.3676, 1.7623]  | 1.0000  | 0.4766    |
| Mobile Weight (grams) [SELECTED]     | -0.4626     | [-0.9405, 0.0320] | 0.9960  | 0.2384    |
| RAM Memory (GB) [SELECTED]           | 0.4268***   | [0.1571, 0.8314]  | 0.9990  | 0.1724    |
| Front Camera (MP) [SELECTED]         | 0.1149      | [-0.0311, 0.3260] | 0.9990  | 0.0945    |
| Max Back Camera MP [SELECTED]        | 0.1823      | [-0.3298, 0.3952] | 0.9940  | 0.4754    |
| Number of Cameras [SELECTED]         | 0.2213***   | [0.0547, 0.4063]  | 1.0000  | 0.0858    |
| Processor Level (encoded) [SELECTED] | -0.0119     | [-0.1102, 0.0968] | 0.6630  | 0.0642    |
| Battery Capacity (mAh) [SELECTED]    | 0.1707      | [-0.4290, 0.6549] | 0.9910  | 0.2848    |
| Screen Size (inches) [SELECTED]      | 0.3286***   | [0.0816, 0.7323]  | 1.0000  | 0.1547    |

Model Information:

- Sample Size: 40
- R<sup>2</sup> Score: 0.8173
- Optimal Alpha: 0.000226
- Features Selected: 9/9

# Lasso Regression Summary: 2021 Q3

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 1.0486***   | [0.5227, 1.6993]  | 1.0000  | 0.3099    |
| Mobile Weight (grams) [SELECTED]     | -0.4757     | [-0.8782, 0.0000] | 0.9580  | 0.2195    |
| RAM Memory (GB) [SELECTED]           | 0.4579***   | [0.2523, 0.7269]  | 0.9980  | 0.1153    |
| Front Camera (MP) [SELECTED]         | 0.0772      | [-0.0679, 0.2473] | 0.9510  | 0.0830    |
| Max Back Camera MP [SELECTED]        | 0.1474      | [-0.1921, 0.3105] | 0.9700  | 0.1142    |
| Number of Cameras [SELECTED]         | 0.2276***   | [0.0855, 0.3686]  | 1.0000  | 0.0717    |
| Processor Level (encoded) [SELECTED] | -0.0029     | [-0.1525, 0.1673] | 0.8170  | 0.0773    |
| Battery Capacity (mAh) [SELECTED]    | 0.2424      | [-0.3325, 0.8109] | 0.7620  | 0.3138    |
| Screen Size (inches) [SELECTED]      | 0.3351***   | [0.1298, 0.6594]  | 0.9950  | 0.1374    |

Model Information:

- Sample Size: 43
- R<sup>2</sup> Score: 0.8548
- Optimal Alpha: 0.004321
- Features Selected: 9/9

# Lasso Regression Summary: 2021 Q4

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI             | P-Value | Std Error |
|-----------------------------------|-------------|--------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED] | 1.1417***   | [0.6826, 1.6215]   | 1.0000  | 0.2392    |
| Mobile Weight (grams) [SELECTED]  | -0.5225***  | [-0.8369, -0.0767] | 0.9800  | 0.1855    |
| RAM Memory (GB) [SELECTED]        | 0.4952***   | [0.2898, 0.7258]   | 1.0000  | 0.0998    |
| Front Camera (MP) [SELECTED]      | 0.0464      | [-0.0822, 0.1989]  | 0.9480  | 0.0705    |
| Max Back Camera MP [SELECTED]     | 0.1386      | [-0.2821, 0.3253]  | 0.9760  | 0.1379    |
| Number of Cameras [SELECTED]      | 0.1966***   | [0.0600, 0.3420]   | 0.9970  | 0.0688    |
| Processor Level (encoded)         | 0.0137      | [-0.1413, 0.1947]  | 0.8730  | 0.0755    |
| Battery Capacity (mAh) [SELECTED] | 0.4444      | [0.0000, 0.7543]   | 0.9310  | 0.2052    |
| Screen Size (inches) [SELECTED]   | 0.2269      | [0.0000, 0.4720]   | 0.9740  | 0.1134    |

Model Information:

- Sample Size: 47
- R<sup>2</sup> Score: 0.8597
- Optimal Alpha: 0.002201
- Features Selected: 8/9



Lasso Regression Summary: 2022 Q1

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI             | P-Value | Std Error |
|--------------------------------------|-------------|--------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 1.1943***   | [0.8299, 1.6542]   | 1.0000  | 0.2049    |
| Mobile Weight (grams) [SELECTED]     | -0.4820***  | [-0.6835, -0.2336] | 1.0000  | 0.1139    |
| RAM Memory (GB) [SELECTED]           | 0.5988***   | [0.3761, 0.8305]   | 1.0000  | 0.1158    |
| Front Camera (MP) [SELECTED]         | 0.0438      | [-0.0969, 0.1983]  | 0.9910  | 0.0733    |
| Max Back Camera MP [SELECTED]        | 0.1796      | [-0.0891, 0.3646]  | 0.9960  | 0.1223    |
| Number of Cameras [SELECTED]         | 0.2175***   | [0.0765, 0.3879]   | 1.0000  | 0.0776    |
| Processor Level (encoded) [SELECTED] | 0.0097      | [-0.1174, 0.1543]  | 0.9420  | 0.0646    |
| Battery Capacity (mAh) [SELECTED]    | 0.6175***   | [0.2383, 0.9277]   | 0.9990  | 0.1750    |
| Screen Size (inches) [SELECTED]      | -0.0158     | [-0.2607, 0.2412]  | 0.9730  | 0.1247    |

Model Information:

- Sample Size: 61
- R² Score: 0.7111
- Optimal Alpha: 0.000264
- Features Selected: 9/9

| Feature                      | Coefficient | 95% CI           | P-Value | Std Error |
|------------------------------|-------------|------------------|---------|-----------|
| iOS (1) vs Android (0)       | 0.0363      | [0.0000, 0.2108] | 0.2110  | 0.0719    |
| Mobile Weight (grams)        | 0.0060      | [0.0000, 0.1098] | 0.0500  | 0.0287    |
| RAM Memory (GB)              | 0.0053      | [0.0000, 0.1251] | 0.0380  | 0.0270    |
| Front Camera (MP)            | 0.0062      | [0.0000, 0.1264] | 0.0500  | 0.0288    |
| Max Back Camera MP           | 0.0000      | [0.0000, 0.0000] | 0.0010  | 0.0011    |
| Number of Cameras [SELECTED] | 0.1880      | [0.0000, 0.3306] | 0.8420  | 0.0957    |
| Processor Level (encoded)    | 0.0564      | [0.0000, 0.2718] | 0.2830  | 0.0955    |
| Battery Capacity (mAh)       | 0.0003      | [0.0000, 0.0000] | 0.0020  | 0.0067    |
| Screen Size (inches)         | 0.0000      | [0.0000, 0.0000] | 0.0000  | 0.0000    |

Model Information:

- Sample Size: 67
- R² Score: 0.0752
- Optimal Alpha: 0.162286
- Features Selected: 1/9

Lasso Regression Summary: 2022 Q3

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.6993***   | [0.3388, 1.1951]  | 0.9990  | 0.2221    |
| Mobile Weight (grams)                | -0.0355     | [-0.4646, 0.1951] | 0.2840  | 0.1396    |
| RAM Memory (GB) [SELECTED]           | 0.4096***   | [0.1885, 0.6185]  | 0.9990  | 0.1163    |
| Front Camera (MP) [SELECTED]         | 0.0376      | [-0.0888, 0.1773] | 0.8060  | 0.0678    |
| Max Back Camera MP [SELECTED]        | 0.0955      | [0.0000, 0.2317]  | 0.8060  | 0.0749    |
| Number of Cameras [SELECTED]         | 0.1134      | [-0.0161, 0.2538] | 0.9880  | 0.0695    |
| Processor Level (encoded) [SELECTED] | 0.1238      | [0.0000, 0.2382]  | 0.9140  | 0.0716    |
| Battery Capacity (mAh) [SELECTED]    | 0.3496      | [0.0000, 0.8169]  | 0.6900  | 0.2809    |
| Screen Size (inches) [SELECTED]      | -0.2379     | [-0.5440, 0.0000] | 0.7420  | 0.1878    |

Model Information:

- Sample Size: 71
- R² Score: 0.5337
- Optimal Alpha: 0.015476
- Features Selected: 8/9

Lasso Regression Summary: 2022 Q4

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.6458***   | [0.4654, 0.8620]  | 1.0000  | 0.0970    |
| Mobile Weight (grams) [SELECTED]     | 0.0149      | [-0.2290, 0.2045] | 0.5910  | 0.0963    |
| RAM Memory (GB) [SELECTED]           | 0.4332***   | [0.2598, 0.6312]  | 1.0000  | 0.0928    |
| Front Camera (MP) [SELECTED]         | 0.0881      | [-0.0078, 0.2001] | 0.9500  | 0.0575    |
| Max Back Camera MP                   | -0.0020     | [-0.1714, 0.1057] | 0.6780  | 0.0671    |
| Number of Cameras [SELECTED]         | -0.0489     | [-0.1607, 0.0609] | 0.7500  | 0.0576    |
| Processor Level (encoded) [SELECTED] | 0.1310**    | [0.0125, 0.2884]  | 0.9830  | 0.0665    |
| Battery Capacity (mAh) [SELECTED]    | 0.2159      | [0.0000, 0.4848]  | 0.9520  | 0.1177    |
| Screen Size (inches)                 | -0.0436     | [-0.2665, 0.1329] | 0.5130  | 0.1001    |

Model Information:

- Sample Size: 86
- R² Score: 0.5740
- Optimal Alpha: 0.010247
- Features Selected: 7/9

Lasso Regression Summary: 2023 Q1

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.5808***   | [0.4453, 0.7553]  | 1.0000  | 0.0781    |
| Mobile Weight (grams) [SELECTED]     | 0.1069      | [-0.0423, 0.2575] | 0.8800  | 0.0877    |
| RAM Memory (GB) [SELECTED]           | 0.4121***   | [0.2483, 0.5841]  | 1.0000  | 0.0855    |
| Front Camera (MP) [SELECTED]         | 0.1482      | [0.0000, 0.2915]  | 0.9750  | 0.0722    |
| Max Back Camera MP [SELECTED]        | 0.0213      | [-0.1242, 0.1563] | 0.7480  | 0.0675    |
| Number of Cameras [SELECTED]         | -0.0889     | [-0.1863, 0.0192] | 0.9050  | 0.0558    |
| Processor Level (encoded) [SELECTED] | 0.0853      | [-0.0106, 0.1924] | 0.9580  | 0.0531    |
| Battery Capacity (mAh) [SELECTED]    | 0.0686      | [-0.1022, 0.3034] | 0.7120  | 0.0980    |
| Screen Size (inches) [SELECTED]      | 0.0305      | [-0.1826, 0.2184] | 0.5980  | 0.0915    |

Model Information:

- Sample Size: 92
- R² Score: 0.5713
- Optimal Alpha: 0.009505
- Features Selected: 9/9

Lasso Regression Summary: 2023 Q2

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED] | 0.4616***   | [0.3340, 0.6131]  | 1.0000  | 0.0725    |
| Mobile Weight (grams) [SELECTED]  | 0.0904      | [-0.0691, 0.2038] | 0.9840  | 0.0623    |
| RAM Memory (GB) [SELECTED]        | 0.3575***   | [0.2089, 0.5098]  | 0.9980  | 0.0767    |
| Front Camera (MP) [SELECTED]      | 0.1115      | [0.0000, 0.2547]  | 0.8490  | 0.0676    |
| Max Back Camera MP                | 0.0407      | [0.0000, 0.1719]  | 0.4090  | 0.0566    |
| Number of Cameras                 | -0.0098     | [-0.1522, 0.0105] | 0.1170  | 0.0393    |
| Processor Level (encoded)         | 0.0214      | [-0.0685, 0.1232] | 0.4560  | 0.0464    |
| Battery Capacity (mAh)            | 0.0160      | [0.0000, 0.2031]  | 0.1070  | 0.0563    |
| Screen Size (inches)              | 0.0138      | [0.0000, 0.1928]  | 0.0990  | 0.0492    |

Model Information:

- Sample Size: 99
- R² Score: 0.4452
- Optimal Alpha: 0.040742
- Features Selected: 4/9

Lasso Regression Summary: 2023 Q3

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                           | Coefficient | 95% CI            | P-Value | Std Error |
|-----------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED] | 0.4816***   | [0.3463, 0.6177]  | 1.0000  | 0.0672    |
| Mobile Weight (grams) [SELECTED]  | 0.0705      | [-0.0507, 0.2003] | 0.9350  | 0.0640    |
| RAM Memory (GB) [SELECTED]        | 0.3837***   | [0.2466, 0.5349]  | 1.0000  | 0.0736    |
| Front Camera (MP) [SELECTED]      | 0.1181      | [0.0000, 0.2301]  | 0.9380  | 0.0562    |
| Max Back Camera MP                | 0.0406      | [0.0000, 0.1684]  | 0.4830  | 0.0541    |
| Number of Cameras                 | -0.0105     | [-0.1260, 0.0474] | 0.2880  | 0.0404    |
| Processor Level (encoded)         | 0.0040      | [-0.0967, 0.0970] | 0.5430  | 0.0471    |
| Battery Capacity (mAh)            | 0.0021      | [-0.1450, 0.1362] | 0.1620  | 0.0543    |
| Screen Size (inches)              | 0.0393      | [0.0000, 0.2134]  | 0.3440  | 0.0650    |

Model Information:

- Sample Size: 102
- R² Score: 0.5155
- Optimal Alpha: 0.026714
- Features Selected: 4/9

Lasso Regression Summary: 2023 Q4

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.5256***   | [0.4124, 0.6664]  | 1.0000  | 0.0621    |
| Mobile Weight (grams) [SELECTED]     | 0.0521      | [-0.1019, 0.1783] | 0.7720  | 0.0667    |
| RAM Memory (GB) [SELECTED]           | 0.3903***   | [0.2604, 0.5293]  | 1.0000  | 0.0683    |
| Front Camera (MP) [SELECTED]         | 0.1541***   | [0.0316, 0.2705]  | 0.9900  | 0.0609    |
| Max Back Camera MP [SELECTED]        | 0.0541      | [-0.0633, 0.1844] | 0.8600  | 0.0579    |
| Number of Cameras [SELECTED]         | -0.0813     | [-0.1582, 0.0000] | 0.9460  | 0.0425    |
| Processor Level (encoded) [SELECTED] | 0.0147      | [-0.0847, 0.1185] | 0.8440  | 0.0509    |
| Battery Capacity (mAh)               | 0.0276      | [-0.1222, 0.1899] | 0.6510  | 0.0731    |
| Screen Size (inches) [SELECTED]      | 0.0375      | [-0.1100, 0.1904] | 0.6510  | 0.0691    |

Model Information:

- Sample Size: 117
- R² Score: 0.5587
- Optimal Alpha: 0.008342
- Features Selected: 8/9



Lasso Regression Summary: 2024 Q1

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI             | P-Value | Std Error |
|--------------------------------------|-------------|--------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.4733***   | [0.3509, 0.6076]   | 1.0000  | 0.0643    |
| Mobile Weight (grams) [SELECTED]     | 0.0580      | [-0.0944, 0.1912]  | 0.7870  | 0.0682    |
| RAM Memory (GB) [SELECTED]           | 0.4122***   | [0.2692, 0.5596]   | 1.0000  | 0.0730    |
| Front Camera (MP) [SELECTED]         | 0.1195      | [0.0000, 0.2371]   | 0.9630  | 0.0626    |
| Max Back Camera MP [SELECTED]        | 0.0363      | [-0.0884, 0.1617]  | 0.8250  | 0.0576    |
| Number of Cameras [SELECTED]         | -0.0973***  | [-0.1741, -0.0004] | 0.9780  | 0.0412    |
| Processor Level (encoded) [SELECTED] | -0.0232     | [-0.1185, 0.0773]  | 0.8610  | 0.0517    |
| Battery Capacity (mAh) [SELECTED]    | 0.0568      | [-0.0981, 0.2231]  | 0.7720  | 0.0773    |
| Screen Size (inches)                 | 0.0043      | [-0.1496, 0.1574]  | 0.6040  | 0.0724    |

Model Information:

- Sample Size: 121
- R² Score: 0.5062
- Optimal Alpha: 0.007584
- Features Selected: 8/9

Lasso Regression Summary: 2024 Q2

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.4873***   | [0.3552, 0.6290]  | 1.0000  | 0.0696    |
| Mobile Weight (grams) [SELECTED]     | 0.0305      | [-0.1403, 0.1594] | 0.8270  | 0.0746    |
| RAM Memory (GB) [SELECTED]           | 0.3786***   | [0.2211, 0.5235]  | 1.0000  | 0.0769    |
| Front Camera (MP) [SELECTED]         | 0.1654***   | [0.0481, 0.2943]  | 1.0000  | 0.0645    |
| Max Back Camera MP [SELECTED]        | 0.0730      | [-0.0451, 0.1877] | 0.9530  | 0.0571    |
| Number of Cameras [SELECTED]         | -0.0695     | [-0.1433, 0.0142] | 0.9660  | 0.0434    |
| Processor Level (encoded) [SELECTED] | -0.0266     | [-0.1246, 0.0684] | 0.9310  | 0.0502    |
| Battery Capacity (mAh) [SELECTED]    | 0.0712      | [-0.0833, 0.2456] | 0.8790  | 0.0835    |
| Screen Size (inches) [SELECTED]      | 0.0332      | [-0.1217, 0.2059] | 0.8090  | 0.0835    |

Model Information:

- Sample Size: 129
- R² Score: 0.4984
- Optimal Alpha: 0.004065
- Features Selected: 9/9

Lasso Regression Summary: 2024 Q3

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.4773***   | [0.3613, 0.5960]  | 1.0000  | 0.0620    |
| Mobile Weight (grams) [SELECTED]     | 0.0433      | [-0.0916, 0.1644] | 0.7350  | 0.0638    |
| RAM Memory (GB) [SELECTED]           | 0.3736***   | [0.2384, 0.5228]  | 1.0000  | 0.0727    |
| Front Camera (MP) [SELECTED]         | 0.1642***   | [0.0210, 0.2827]  | 0.9920  | 0.0651    |
| Max Back Camera MP [SELECTED]        | 0.0641      | [0.0000, 0.1676]  | 0.8830  | 0.0531    |
| Number of Cameras [SELECTED]         | -0.0557     | [-0.1259, 0.0290] | 0.8650  | 0.0417    |
| Processor Level (encoded) [SELECTED] | -0.0196     | [-0.1203, 0.0748] | 0.8350  | 0.0489    |
| Battery Capacity (mAh) [SELECTED]    | 0.0448      | [-0.1165, 0.2163] | 0.6760  | 0.0773    |
| Screen Size (inches) [SELECTED]      | 0.0336      | [-0.1062, 0.1680] | 0.6070  | 0.0643    |

Model Information:

- Sample Size: 128
- R² Score: 0.4950
- Optimal Alpha: 0.009347
- Features Selected: 9/9

Lasso Regression Summary: 2024 Q4

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.4980***   | [0.3677, 0.6408]  | 1.0000  | 0.0676    |
| Mobile Weight (grams) [SELECTED]     | 0.0299      | [-0.1411, 0.1704] | 0.9130  | 0.0777    |
| RAM Memory (GB) [SELECTED]           | 0.3840***   | [0.2360, 0.5351]  | 1.0000  | 0.0763    |
| Front Camera (MP) [SELECTED]         | 0.1402***   | [0.0156, 0.2614]  | 0.9950  | 0.0631    |
| Max Back Camera MP [SELECTED]        | 0.0502      | [-0.0599, 0.1583] | 0.9470  | 0.0574    |
| Number of Cameras [SELECTED]         | -0.0916     | [-0.1693, 0.0000] | 0.9960  | 0.0423    |
| Processor Level (encoded) [SELECTED] | -0.0547     | [-0.1512, 0.0370] | 0.9820  | 0.0482    |
| Battery Capacity (mAh) [SELECTED]    | 0.1033      | [-0.0378, 0.2794] | 0.9670  | 0.0787    |
| Screen Size (inches)                 | 0.0066      | [-0.1433, 0.1768] | 0.8970  | 0.0829    |

Model Information:

- Sample Size: 140
- R² Score: 0.5223
- Optimal Alpha: 0.001761
- Features Selected: 8/9

Lasso Regression Summary: 2025 Q1

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.4770***   | [0.3400, 0.6348]  | 1.0000  | 0.0748    |
| Mobile Weight (grams)                | -0.0134     | [-0.1804, 0.1312] | 0.8220  | 0.0755    |
| RAM Memory (GB) [SELECTED]           | 0.3913***   | [0.2554, 0.5248]  | 1.0000  | 0.0705    |
| Front Camera (MP) [SELECTED]         | 0.1433***   | [0.0171, 0.2710]  | 0.9930  | 0.0690    |
| Max Back Camera MP [SELECTED]        | 0.0722      | [-0.0389, 0.1942] | 0.9550  | 0.0595    |
| Number of Cameras [SELECTED]         | -0.0639     | [-0.1506, 0.0266] | 0.9730  | 0.0435    |
| Processor Level (encoded) [SELECTED] | -0.0546     | [-0.1432, 0.0278] | 0.9500  | 0.0452    |
| Battery Capacity (mAh) [SELECTED]    | 0.1144      | [-0.0394, 0.2761] | 0.9590  | 0.0825    |
| Screen Size (inches) [SELECTED]      | 0.0485      | [-0.1076, 0.2349] | 0.8690  | 0.0864    |

Model Information:

- Sample Size: 146
- R² Score: 0.5258
- Optimal Alpha: 0.003013
- Features Selected: 8/9

Lasso Regression Summary: 2025 Q2

[SELECTED] = Feature selected by Lasso | Significance: \*\*\* |coef|>2\*SE, \*\* |coef|>1.96\*SE, \* |coef|>1.645\*SE

| Feature                              | Coefficient | 95% CI            | P-Value | Std Error |
|--------------------------------------|-------------|-------------------|---------|-----------|
| iOS (1) vs Android (0) [SELECTED]    | 0.5316***   | [0.4023, 0.6698]  | 1.0000  | 0.0692    |
| Mobile Weight (grams) [SELECTED]     | -0.1095     | [-0.2778, 0.0187] | 0.9950  | 0.0763    |
| RAM Memory (GB) [SELECTED]           | 0.3930***   | [0.2206, 0.5521]  | 1.0000  | 0.0845    |
| Front Camera (MP) [SELECTED]         | 0.1257      | [-0.0109, 0.2592] | 0.9990  | 0.0670    |
| Max Back Camera MP [SELECTED]        | 0.0719      | [-0.0552, 0.1940] | 0.9970  | 0.0625    |
| Number of Cameras [SELECTED]         | -0.0500     | [-0.1229, 0.0374] | 0.9960  | 0.0414    |
| Processor Level (encoded) [SELECTED] | -0.0659     | [-0.1465, 0.0221] | 0.9960  | 0.0427    |
| Battery Capacity (mAh) [SELECTED]    | 0.1477      | [-0.0103, 0.3136] | 0.9960  | 0.0809    |
| Screen Size (inches) [SELECTED]      | 0.1691**    | [0.0119, 0.3346]  | 0.9960  | 0.0855    |

Model Information:

- Sample Size: 133
- R² Score: 0.5776
- Optimal Alpha: 0.000252
- Features Selected: 9/9